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From Immersion to Loyalty:
Virtual Interaction and Brand Trust in PICO VR Brand Relationships

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Abstract: With the rapid development of virtual reality (VR) and metaverse technologies, immersive brand experience has become a critical pathway through which brands establish deeper relationships with consumers. However, the mechanism by which immersive experience is transformed into brand trust and brand loyalty remains insufficiently explained, particularly in consumer VR settings where interaction, presence, and community participation are closely intertwined.

This study focuses on PICO VR users and constructs a structural model linking immersive virtual brand experience (IVBE), virtual interaction (VI), brand trust (BT), and brand loyalty (BL). Based on 505 valid questionnaire responses, this paper applies reliability testing, confirmatory factor analysis, structural equation modeling, and mediation analysis to examine the direct and indirect effects among the four constructs.

The empirical results show that IVBE has significant positive effects on VI, BT, and BL. VI also positively affects both BT and BL, while BT further promotes BL. Mediation testing confirms that VI partially mediates the effects of IVBE on both BT and BL. These findings indicate that immersive experience alone is not sufficient for long-term brand relationship building; rather, virtual interaction functions as a key psychological and behavioral bridge that converts immersive perception into trust and loyalty.

Keywords: Immersive Virtual Brand Experience; Virtual Interaction; Brand Trust; Brand Loyalty; Structural Equation Modeling; PICO VR; Metaverse

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1 Introduction

Virtual reality has moved from a technology-centered device market toward an experience-centered brand environment. In metaverse-related consumption contexts, users do not merely evaluate products through functional attributes such as hardware performance or price. They also form brand attitudes through sensory immersion, situational presence, smooth interaction, social participation, and emotional attachment. These changes require brand researchers to explain how immersive experience is converted into stable trust and loyalty.

PICO VR provides a meaningful context for examining this mechanism. As an important VR brand in China, PICO integrates entertainment, fitness, social interaction, virtual cinema, and platform-based content services. These usage scenarios allow users to interact not only with devices but also with virtual environments, digital content, and other users. Therefore, the brand relationship formed in PICO VR is not a simple outcome of product satisfaction; it is also shaped by the quality and depth of virtual interaction.

Existing research has confirmed that brand experience, trust, and loyalty are closely connected (Chaudhuri & Holbrook, 2001; Morgan & Hunt, 1994). In immersive environments, however, the pathway may become more complex because user participation occurs through real-time interaction and virtual presence. The Stimulus-Organism-Response (S-O-R) framework suggests that external experiential stimuli influence users internal psychological states and then shape behavioral responses. In this paper, IVBE is treated as the experiential stimulus, VI and BT as key psychological and relational mechanisms, and BL as the behavioral outcome.

Accordingly, this study aims to answer three questions. First, does IVBE significantly enhance BT and BL among PICO VR users? Second, does VI strengthen the relationship between IVBE and brand relationship outcomes? Third, does BT remain an important antecedent of BL in metaverse-based brand environments? By addressing these questions through survey data and structural equation modeling, the paper provides a focused empirical explanation of how immersive brand relationships are formed.

The contribution of this study lies in three aspects. Theoretically, it extends brand relationship research into immersive VR contexts by integrating IVBE, VI, BT, and BL into a unified model. Methodologically, it validates the measurement structure through CFA and SEM using a sample of 505 respondents. Practically, it offers managerial insights for VR brands seeking to improve user retention through interaction design, trust-building mechanisms, and immersive content strategies.

2 Theoretical and literature foundation

2.1 Immersive Virtual Brand Experience

Immersive virtual brand experience refers to users multi-sensory and psychologically engaging experience with a brand in a virtual environment. Compared with traditional brand experience, IVBE emphasizes presence, realism, interaction fluency, content involvement, and technological reliability. These dimensions allow users to perceive the brand as an experiential environment rather than merely as a product provider (Schmitt, 1999; Pine & Gilmore, 1999; Mingione et al., 2024).

In VR contexts, immersion can increase perceived vividness and reduce the psychological distance between users and brands. When a user feels that the virtual scene is realistic, interactive, and controllable, the brand can more easily generate emotional resonance and cognitive recognition. Therefore, IVBE is expected to act as the initial driver of user trust, interaction, and loyalty.

2.2 Virtual Interaction as a Relational Mechanism

Virtual interaction refers to users active engagement with virtual environments, brand-related content, platform communities, avatars, and other users. It includes behavioral participation, community involvement, emotional response, and immersive communication. In metaverse environments, VI is not limited to technical operation; it also reflects users sense of agency, belonging, and co-presence (Lee & Cho, 2022; Park & Kim, 2022).

The S-O-R framework provides a theoretical basis for understanding VI as a mediating mechanism. Immersive experience provides the stimulus, while interaction functions as an organismic response that deepens user involvement and emotional connection. Through repeated interaction, users obtain feedback, social identity, and perceived control, which may strengthen trust in the brand and reinforce loyalty intentions.

2.3 Brand Trust and Brand Loyalty in VR Contexts

Brand trust refers to consumers confidence that a brand is competent, sincere, reliable, and able to fulfill its promises. In VR environments, trust is particularly important because users often face uncertainty related to technological performance, content quality, privacy, and service reliability (Delgado-Ballester & Munuera-Aleman, 2001; Hegner & Jevons, 2016).

Brand loyalty includes repurchase intention, recommendation willingness, continued use, and sustained participation in brand-related communities. In immersive platforms, loyalty is not only reflected in product choice but also in continuous presence, content engagement, and community attachment. Prior studies suggest that trust is a central antecedent of loyalty because it reduces perceived risk and increases relationship commitment (Chaudhuri & Holbrook, 2001; Huang, 2017).

2.4 Research Hypotheses

Based on the above theoretical discussion, this paper proposes that IVBE directly promotes BT and BL, while also increasing VI. VI is expected to enhance BT and BL because interactive participation strengthens psychological engagement and relational attachment. Finally, BT is expected to promote BL as a core trust-loyalty pathway in digital brand relationships.

H1: Immersive Virtual Brand Experience has a significant positive effect on Brand Trust.

H2: Immersive Virtual Brand Experience has a significant positive effect on Brand Loyalty.

H3: Immersive Virtual Brand Experience has a significant positive effect on Virtual Interaction.

H4: Virtual Interaction has a significant positive effect on Brand Trust.

H5: Virtual Interaction has a significant positive effect on Brand Loyalty.

H6: Brand Trust has a significant positive effect on Brand Loyalty.

H7a: Virtual Interaction partially mediates the relationship between Immersive Virtual Brand Experience and Brand Trust.

H7b: Virtual Interaction partially mediates the relationship between Immersive Virtual Brand Experience and Brand Loyalty.

3 Research method

3.1 Research Design and Measurement

This study adopts a quantitative survey design to test the structural relationships among IVBE, VI, BT, and BL. All constructs were measured using a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The final scale included 16 items: five for IVBE, four for VI, three for BT, and four for BL. The measurement design was adapted from established studies and adjusted to the PICO VR usage context.

Table 1. Constructs and Measurement Focus

Construct	Items	Measurement Focus	Main References
IVBE	5	Sensory immersion, situational presence, interaction fluency, content engagement, technological reliability	Mingione et al. (2024); Fokides (2023); Lee & Cho (2022)
VI	4	Behavioral interaction, community engagement, emotional response, immersive communication	Lee & Cho (2022); Park & Kim (2022); Payal et al. (2024)
BT	3	Technological competence, benevolence toward users, integrity and promise fulfillment	Delgado-Ballester & Munuera-Aleman (2001); Hegner & Jevons (2016)
BL	4	Repurchase intention, recommendation, virtual presence loyalty, community loyalty	Chaudhuri & Holbrook (2001); Huang (2017); Kaur et al. (2019)

3.2 Data Collection and Sample

The questionnaire was distributed through an online survey platform between July 2, 2025 and August 20, 2025. Convenience sampling and snowball sampling were used to reach respondents with PICO VR usage experience or familiarity with VR-based immersive consumption scenarios. The questionnaire introduction explained the research purpose, anonymity principle, and response requirements.

A total of 512 questionnaires were collected. After excluding invalid responses with extremely short completion times or inconsistent answering patterns, 505 valid questionnaires were retained for analysis. The sample size satisfies the common

SEM requirement for medium-complexity models and provides adequate statistical power for estimating latent-variable relationships (Hair et al., 2010).

The sample was relatively balanced in gender, with 232 male respondents (45.9%) and 273 female respondents (54.1%). Most respondents were between 18 and 40 years old, accounting for 67.3% of the sample. In terms of education, 53.6% held a bachelor degree or above. Regarding usage behavior, 63.0% reported using PICO VR for 4 to 21 hours per week, suggesting that the sample included a substantial proportion of medium-frequency users.

3.3 Analytical Procedure

SPSS 26.0 and AMOS 26.0 were used for statistical analysis. First, descriptive statistics were conducted to examine the sample profile. Second, Cronbachs alpha and KMO values were calculated to test internal consistency and sampling adequacy. Third, confirmatory factor analysis was performed to evaluate factor loadings, composite reliability (CR), and average variance extracted (AVE). Finally, structural equation modeling was used to test the hypotheses, and mediation analysis was conducted to examine the indirect effects of VI.

4 Data analysis and results

4.1 Descriptive Results

The descriptive results indicate that PICO VR is used across multiple immersive scenarios, especially gaming and entertainment, virtual cinema, fitness, and social interaction. The concentration of respondents in medium-frequency usage groups suggests that immersive VR experiences have become part of regular entertainment and leisure routines for many users. This provides a suitable empirical basis for examining brand trust and loyalty formation in an immersive platform context.

The demographic distribution also suggests that VR adoption is not limited to a narrow early-adopter group. Younger users remain the core segment, but respondents above 40 also formed a meaningful share of the sample. This diversity strengthens the applicability of the findings to broader consumer VR markets.

4.2 Reliability and Validity Analysis

The measurement model demonstrated good reliability and validity. Cronbachs alpha values for all constructs exceeded 0.80, indicating strong internal consistency. KMO values were above the recommended threshold, showing that the data were suitable for factor analysis. CFA results further showed that standardized factor loadings were all above 0.70, while CR values exceeded 0.80 and AVE values exceeded 0.50. These results confirm satisfactory convergent validity and construct reliability.

Table 2. Reliability and Validity Results

Construct	Cronbachs alpha	KMO	CR	AVE	Std. Loading Range
IVBE	0.881	0.884	0.881	0.597	0.762-0.797
VI	0.859	0.823	0.859	0.604	0.772-0.783
BT	0.820	0.714	0.821	0.606	0.719-0.812
BL	0.873	0.833	0.874	0.634	0.751-0.831

4.3 Structural Equation Modeling

The structural model showed an excellent overall fit: CMIN/DF = 1.027, RMR = 0.030, RMSEA = 0.007, GFI = 0.977, NFI = 0.976, CFI = 0.999, and AGFI = 0.968. All indices met or exceeded commonly accepted standards, indicating that the proposed model was highly consistent with the observed data.

The path results support all direct-effect hypotheses. IVBE significantly influenced BT (standardized coefficient = 0.240), BL (0.212), and VI (0.505). VI significantly influenced BT (0.360) and BL (0.223), while BT significantly influenced BL (0.284). Among the paths, IVBE -> VI was the strongest, suggesting that immersive experience is especially effective in stimulating interaction in the PICO VR context.

Table 3. Structural Path and Mediation Results

Hyp.	Path	Coefficient / Effect	Significance or CI	Decision

H1	IVBE -> BT	Std. = 0.240	$p < 0.001$	Supported
H2	IVBE -> BL	Std. = 0.212	$p < 0.001$	Supported
H3	IVBE -> VI	Std. = 0.505	$p < 0.001$	Supported
H4	VI -> BT	Std. = 0.360	$p < 0.001$	Supported
H5	VI -> BL	Std. = 0.223	$p < 0.001$	Supported
H6	BT -> BL	Std. = 0.284	$p < 0.001$	Supported
H7a	IVBE -> VI -> BT	Indirect = 0.1359	95% CI [0.0928, 0.1829]	Supported
H7b	IVBE -> VI -> BL	Indirect = 0.1355	95% CI [0.0917, 0.1832]	Supported

4.4 Discussion of Findings

The results reveal a clear path from immersion to loyalty. IVBE not only directly enhances BT and BL but also strongly increases VI. This means that immersive perception is valuable because it motivates users to participate, communicate, and remain active within the virtual environment. For PICO VR, improving the realism, smoothness, and richness of immersive scenarios can therefore increase the likelihood of user interaction.

VI plays a partial mediating role in both the IVBE -> BT path and the IVBE -> BL path. This finding suggests that users do not develop trust and loyalty merely because a virtual experience is vivid. Trust and loyalty are strengthened when users can interact smoothly, obtain emotional feedback, and feel a sense of belonging in the platform environment. Interaction design is therefore a key managerial lever for turning experiential quality into relationship quality.

The positive effect of BT on BL confirms that trust remains central in immersive digital environments. Even when experiences are novel and entertaining, users still rely on perceptions of technological competence, sincerity, and promise fulfillment when deciding whether to continue using and recommending a brand. VR brands should therefore integrate experience innovation with stable service quality and transparent trust-building practices.

5 Conclusion

This study examined how immersive virtual brand experience influences brand trust and brand loyalty among PICO VR users. Based on 505 valid survey responses and structural equation modeling, the results support a relational pathway in which IVBE promotes VI, BT, and BL; VI further strengthens BT and BL; and BT positively contributes to BL. The mediation results confirm that VI functions as an important bridge between immersive experience and brand relationship outcomes.

Theoretically, this paper extends brand relationship research by highlighting the mediating role of VI in immersive VR environments. It shows that the value of IVBE lies not only in sensory stimulation but also in its capacity to activate interaction, participation, and psychological connection. The model therefore enriches the application of the S-O-R framework in metaverse-based brand research.

Practically, the findings suggest that VR brands should not rely solely on hardware novelty or content quantity. To cultivate sustainable loyalty, brands need to design interactive scenarios that support real-time feedback, community participation, avatar-based communication, and user co-creation. At the same time, trust-building mechanisms such as stable product performance, responsive service, privacy protection, and transparent communication should be strengthened.

This study also has limitations. The data are cross-sectional and focus on one VR brand, which limits causal inference and generalizability. Future research may adopt longitudinal or experimental designs, compare multiple VR brands, and introduce moderating variables such as technology readiness, perceived value, social presence, and community identification. Such extensions would further clarify how immersive brand relationships evolve over time.

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